CY34/CI34

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RAMAIAH Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

MAKE UP EXAMINATIONS – JUNE 2023

Program	:	**B.E. - CSE (Cyber Security) / CSE (Artificial Intelligence and Machine Learning)**	**Semester**	:	**III**
Course Name	:	**Computer Organization and Architecture**	**Max. Marks**	:	**100**
Course Code	:	**CY34 / CI34**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1.	a)	Differentiate between combinational and sequential circuits. Describe the working of full adder with neat circuit diagram and truth table.	CO1	(08)
	b)	Minimize the following expression using laws of Boolean algebra: $(A+B+C)\ (A+\bar{B}+C)\ (A+B+\bar{C})$.	CO1	(04)
	c)	Describe the working of JK- flipflop with neat circuit diagram and truth table.	CO1	(08)
2.	a)	Describe the three different classes of computing applications and their characteristics.	CO1	(08)
	b)	With neat circuit diagram and truth table, describe 4:1 multiplexer.	CO1	(08)
	c)	Consider a logic circuit with three inputs A, B, and C. Output Y is high for the following conditions: i. B and C are true ii. A and C are false iii. A, B, and C are true iv. A, B and C are false. Minimize the function Y using Law's of Boolean Algebra.	CO1	(04)

UNIT - II

3	a)	Describe any Five data transfer instructions used by MIPS architecture.	CO2	(06)
	b)	With the help of circuit diagram discuss First version of the multiplication hardware.	CO4	(06)
	c)	List and explain different addressing modes with example.	CO2	(08)
4	a)	Multiply 4x3 using multiplication algorithm.	CO4	(04)
	b)	With an Example, explain 2 types of MIPS instruction formats.	CO2	(06)
	c)	With the help of circuit diagram discuss an improved version of the division hardware.	CO4	(10)

UNIT - III

5.	a)	Discuss the direct mapping and associative cache memory mapping functions with suitable examples.	CO3	(08)
	b)	Describe different types of Read only memories.	CO3	(06)

	c)	Explain the basic structure of the memory hierarchy along with a neat diagram.	CO3	(06)
6.	a)	Illustrate the characteristics of virtual memory.	CO3	(10)
	b)	Discuss how to compute the cache performance. Assume the miss rate of an instruction cache is 3% and the miss rate of the data cache is 5%. If a processor has a CPI of 2 without any memory stalls and the miss penalty is 150 cycles for all misses, determine how much faster a processor would run with a perfect cache that never missed. Assume the frequency of all loads and stores is 45%.	CO3	(10)

UNIT- IV

7.	a)	Explain how instruction fetch cycle works along with a neat diagram and also mention the various registers involved in this.	CO4	(08)
	b)	Describe the Hardwired control unit with a neat diagram.	CO4	(06)
	c)	Differentiate between multiprocessor and multicore processor.	CO4	(06)
8.	a)	Describe the process of fetching and executing instructions, using the Datapath. Explain the control sequences and steps followed for execution of an instruction Add, R3, (R2).	CO4	(08)
	b)	Compare and contrast between CISC and RISC-style processors.	CO4	(06)
	c)	Illustrate the functioning of GPU system architecture.	CO4	(06)

UNIT - V

9.	a)	What do you mean by Multithreading? List the difference between Fine Grain multithreading and Coarse grain multithreading.	CO5	(10)
	b)	Classify parallel computers based on Flynn's Taxonomy.	CO5	(10)
10.	a)	Discuss and list the difference between shared memory multiprocessor and message passing multiprocessors in computer architecture.	CO5	(10)
	b)	Differentiate between CPU and GPU.	CO5	(10)

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